

Part 2: Indexing and Evaluation

2.1. Indexing

1. Build inverted index: After having pre-processed the data, you can then create the inverted index.

We changed the way we handle punctuation marks in part 2 of preprocessing. We now keep only hyphens to preserve compound words.

At first, we considered using the row number of each product as its document ID in the inverted index. However, after reviewing the assignment instructions, we realized this approach would not work. The file `validation_labels.csv` uses unique product IDs to determine relevance, so using row numbers would prevent us from correctly matching our search results with the relevance labels in `validation_labels.csv`.

We chose to build the inverted index using only the title and description fields from the product dataset. These fields provide the most useful and consistent text for search. Since `validation_labels.csv` doesn't include other fields like category or brand, and relevance is based on product IDs and titles, indexing title and description keeps our system aligned with the evaluation process.

2. Propose test queries: Define five queries that will be used to evaluate your search engine.

These are the queries we have selected:

q1 = "casual half sleeve polo shirt for men"

q2 = "light blue jeans slim fit"

q3 = "trousers chino casual men"

q4 = "black sports shoes"

q5 = "fancy t-shirt"

We selected our test queries using some terms with high term frequency mixed with other terms that are less frequent. Popular terms reflect real user search behavior and ensure a large enough result set for evaluation. This helps us test the search engine's ability to retrieve and rank relevant documents. Queries with common words produce relevant and irrelevant results, allowing us to evaluate precision, recall, and ranking performance.

2.2. Evaluation

3. You will act as expert judges by establishing the ground truth for each document and query.

- a) **For the test queries you defined in Part 1, Step 2 during indexing, assign a binary relevance label to each document: 1 if the document is relevant to the query, or 0 if it is not.**

To perform this task, we created a new CSV file (student_validation_labels.csv), which follows the same structure as the provided (validation_labels.csv). We analyzed the images (from the url field) of the top 10 retrieved products for each query and manually assigned relevance labels (1 or 0) based on their relevance to the query. For the evaluation metrics, however, we only considered the top 5 documents. This decision prevents obtaining a recall of 100%, which would occur if the same number of products were used for both the relevance labeling and the evaluation.

- b) **Comment on each of the evaluation metrics, stating how they differ, and which information gives each of them. Analyze your results.**

Precision@K measures the proportion of relevant documents among the top K retrieved results. For example, if 7 out of the top 10 retrieved documents are relevant, then $P@10 = 0.7$.

Recall@K measures the proportion of all relevant documents that appear within the top K retrieved results. For example, if there are 20 relevant documents in total and 8 are found in the top 10 results, then $R@10 = 0.4$.

Average Precision@K calculates the average of the precision values obtained each time a relevant document is retrieved, up to position K. It rewards systems that rank relevant documents higher. Unlike Precision@K, which only counts how many relevant documents appear in the top K, AP@K also considers their ranking positions.

The F1-Score@K is the harmonic mean of Precision@K and Recall@K. It provides a single measure that balances both precision and recall.

Mean Average Precision is the mean of the Average Precision scores across all queries. It gives an assessment of the ranking quality of the system. While AP@K is calculated per query, MAP aggregates those values to summarize performance over the entire dataset.

Mean Reciprocal Rank measures the average of the reciprocal ranks of the first relevant document retrieved for each query. It focuses on how early the first relevant result appears in the ranking. For example, if the first relevant document is ranked third, its reciprocal rank is 1/3.

NDCG evaluates the ranking quality based on relevance and document position. It assigns higher scores when highly relevant documents appear near the top and applies a discount factor as the rank increases. Unlike MAP or MRR, NDCG can handle multiple levels of relevance.

The results we obtain for our personalized queries are:

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Query 1: casual half sleev polo shirt men
Precision@5      : 1.0
Recall@5         : 0.5
Avg Precision@5  : 1.0
F1-Score@5       : 0.667
Reciprocal Rank@5 : 1.0
NDCG@5           : 1.0
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Query 2: light blue jean slim fit
Precision@5      : 1.0
Recall@5         : 0.625
Avg Precision@5  : 1.0
F1-Score@5       : 0.769
Reciprocal Rank@5 : 1.0
NDCG@5           : 1.0
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Query 3: trouser chino casual men
Precision@5      : 1.0
Recall@5         : 0.556
Avg Precision@5  : 1.0
F1-Score@5       : 0.714
Reciprocal Rank@5 : 1.0
NDCG@5           : 1.0
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Query 4: black sport shoe
Precision@5      : 0.2
Recall@5         : 0.5
Avg Precision@5  : 1.0
F1-Score@5       : 0.286
Reciprocal Rank@5 : 1.0
NDCG@5           : 0.613
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Query 5: fanci t-shirt
Precision@5      : 0.8
Recall@5         : 1.0
Avg Precision@5  : 0.804
F1-Score@5       : 0.889
Reciprocal Rank@5 : 1.0
NDCG@5           : 0.905
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Overall Performance Summary
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Mean Average Precision (MAP@5): 0.961
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Overall, the retrieval system performs well, achieving a Mean Average Precision of 0.961, indicating that most relevant products are ranked highly across the queries.

For Queries 1–3, the performance is excellent, with $\text{Precision@5} = 1.0$ and $\text{NDCG@5} = 1.0$ in all three cases. This means that all retrieved results are relevant and well ranked. However, recall values between 0.5 and 0.6 suggest that although the top results are accurate, not all relevant items are retrieved within the top five positions. This is due to the system retrieving only the top 5 although we have labeled manually the first 10 products (which contain more relevant products). We cannot retrieve the 10 labeled products, because this would result in a recall of 100% in all the queries.

Query 4 shows a significant drop in performance ($\text{Precision@5} = 0.2$, $\text{F1-Score@5} = 0.286$, $\text{NDCG@5} = 0.613$). This reflects issues in the ranking and filtering process, caused by irrelevant products being matched due to misleading metadata or keywords, as we will discuss later (in c).

Query 5 (“fanci t-shirt”) performs well overall, with $\text{Precision@5} = 0.8$ and $\text{Recall@5} = 1.0$, indicating that most retrieved results are relevant and ranked appropriately.

c) Analyze the current search system and identify its main problems or limitations. For each issue you find, propose possible ways to resolve it. Consider aspects such as retrieval accuracy, ranking quality, handling of different field types, query formulation, and indexing strategies.

The description of some of the shoes products announce offers for other products which contain the words in the query. For example, when checking the JSON description of the product with PID SHOFUH52V5AVNHMN, it states “This season Rockfield has introduced sports shoes for both men and boys.” This description is unrelated to the actual product and only matches the query by coincidence. As a result, Query 4 shows low precision and low numbers of relevant products retrieved (2 only). To address this issue, the retrieval system could implement text preprocessing and filtering steps to remove generic or promotional sentences that are not specific to the product itself.

In addition, we have found that when using the query “black sports shoes”, one of the retrieved results was a pair of sneakers that were entirely yellow with only small black details (as it can be seen in the image below). The product was tagged with the colors (yellow, black), which explains why it appeared in the search results. However, from a user’s perspective, this result is not truly relevant, since the dominant color of

the shoes is yellow rather than black. This highlights a limitation in the retrieval process, where the presence of a keyword in the metadata (in this case, “black”) can lead to false positives when the attribute is not representative of the product’s overall appearance. This issue is harder to address. However, a way to fix this problem could be to incorporate computer vision models to automatically detect the primary color from the product image.



product_id = SHOE7TZFE2TCNPH5 - product_title: Sports Shoes For Women (Black, Yellow)

2.3. Tag of the repository

This report references the notebook in:

https://github.com/IsaacPerez02/SearchEngine-IRWA_2025/releases/tag/IRWA-2025-part-2